

Reliability-Aware Transfer Learning for Robust, Calibrated, and Explainable UAV-Based Aquatic Plant Classification

Abstract—Unmanned Aerial Vehicle (UAV) imagery can support efficient wetland monitoring by recognizing aquatic plants. However, high clean-set accuracy does not guarantee reliable deployment. This study presents a reliability-aware transfer-learning framework for the classification of 14 aquatic plants. This framework collectively considers data integrity, robustness, confidence calibration, selective prediction, explainability, and residual failure behavior. An audit of the original 1,468-image classification dataset revealed complete exact overlap between the supplied training and test partitions. After duplicate removal and near-duplicate-aware grouping, 1,062 unique images were repartitioned into 744 training, 161 validation, and 157 locked test images. Four CNN-based models were compared, with ResNet50 and EfficientNet-B0 selected for reliability evaluation. In the 157-image locked test set, ResNet50 correctly classified 151 images (96.18%) with a macro-F1 of 95.90%. Under strong image corruption, ResNet50 retained 91.52% macro-F1, compared to 83.40% for EfficientNet-B0. Temperature scaling reduced ResNet50’s expected calibration error from 0.103 to 0.030. At approximately 90% selective coverage, it achieved 99.30% selective accuracy while rejecting 5 of its 6 original test errors (83.33%). Grad-CAM, Score-CAM and integrated failure analysis further identified shared and high-confidence residual failure cases.

Index Terms—aquatic plant classification, UAV imagery, transfer learning, robustness, calibration, selective prediction, Grad-CAM, explainable AI

I. INTRODUCTION

Aquatic vegetation plays an important role in wetland ecosystems. This supports biodiversity, habitat structure, and water-system stability. However, conventional field surveys are labor-intensive, time-consuming, and difficult to scale across large or inaccessible water bodies. Unmanned aerial vehicles (UAVs) provide a practical alternative by enabling flexible acquisition of high-resolution imagery for vegetation monitoring. Recent studies have demonstrated the usefulness of UAV-based imaging for aquatic vegetation mapping and recognition. Szabó *et al.* used multispectral UAS imagery with spectral, texture, and surface-height information for aquatic vegetation mapping [1], while Yu *et al.* combined UAV hyperspectral imagery with a transformer-based segmentation model for aquatic plant detection in crab ponds [2]. The AqUavplant dataset supported high-resolution UAV-based classification and segmentation research for aquatic plant species [3].

More recently, Rahman *et al.* introduced an attention-enhanced multi-task framework for simultaneous binary segmentation and 14-class aquatic plant classification from UAV imagery, reporting strong classification performance and

demonstrating the value of joint spatial and semantic learning [4]. These studies show that deep learning can achieve high predictive performance for aquatic vegetation analysis. Nevertheless, strong clean-image accuracy alone does not fully describe whether a classifier is reliable under practical deployment conditions. UAV imagery can be affected by blur, sensor noise, illumination changes, compression artifacts, partial occlusion, and visually complex backgrounds. In addition, neural networks can assign high confidence to incorrect predictions. Modern deep classifiers are known to produce miscalibrated probability estimates, motivating explicit calibration analysis in addition to conventional accuracy metrics [5].

Cross-partition duplication can produce overly optimistic generalization estimates. Therefore, reliable evaluation also depends on the integrity of the dataset [6]. Our dataset audit revealed substantial exact duplication of train–test [7]. The original partition was therefore rejected, and a leakage-controlled group-aware split was constructed after exact- and near-duplicate auditing.

This study consequently treats aquatic plant recognition as a *reliability-aware classification problem* rather than only a model-accuracy comparison. A custom CNN, VGG16, ResNet50, and EfficientNet-B0 are first compared using validation performance, after which the selected models are evaluated on a locked test set. Reliability is then examined under six image corruptions at three severity levels, followed by confidence calibration using temperature scaling. Selective prediction is used to determine whether uncertain predictions can be rejected instead of forcing potentially unreliable labels [8]. Finally, Grad-CAM is employed to examine model attention in correct, uncertain, and erroneous cases [9], and the resulting explanations are integrated with uncertainty and error patterns.

The contributions of this study are fourfold: (1) a leakage-aware reconstruction of the UAV aquatic-plant classification dataset using exact and near-duplicate auditing; (2) a transfer-learning evaluation that extends beyond clean accuracy to corruption robustness; (3) confidence calibration and selective prediction to identify and reject unreliable decisions; and (4) reliability-linked Grad-CAM and Score-CAM analysis with integrated failure analysis for characterizing recurrent, shared, and high-confidence residual errors. Together, these contributions provide a more complete evaluation of UAV-based aquatic plant classifiers for practical environmental monitoring.

II. RELATED WORK

UAV-based remote sensing has proved to be increasingly significant to aquatic vegetation monitoring due to its capacity to gain high-resolution spatial information with flexible and comparatively cheaper cost data acquisition. Previous studies have shown the use of unmanned aerial systems for supporting aquatic vegetation mapping in a wetland context [10]. Recent studies have integrated multispectral and hyperspectral information to achieve more detailed discrimination of aquatic vegetation. Szabó et al. proposed using spectral, texture and surface height features for the task of aquatic vegetation mapping with their study [1], while Yu et al. presented a transformer based semantic segmentation model applied to UAV hyperspectral imagery for the purpose of detecting aquatic plants [2]. An additional AqUavplant dataset was also used to provide with high resolution UAV images for the purposes of aquatic plant classification and segmentation to allow the development of data driven recognition methods [3].

Deep convolutional neural networks and transfer learning strategies have improved the performance of image classification tasks in cases of insufficient availability of task specific datasets. As a specific example, the VGG16 network offered a deep stacked convolutional architecture [11], whereas residual learning in the ResNet framework improved optimization of deep convolutional models by introducing identity connections [12]. The EfficientNet family proposed compound scaling of the network depth, width, and input image resolution to obtain a better trade off between accuracy and computational cost [13]. Those architectures are typically initialized with ImageNet pretrained weights [14] and further adapted to the smaller size of domain specific datasets. In the scope of aquatic plant analysis, Rahman et al. recently proposed an attention enhanced multi task framework to continuously perform segmentation and 14-class classification which achieved a compelling predictive performance and Grad-CAM-based interpretability [4]. Though the mentioned study reported substantially better classification performance, a direct comparison with the current results should be made with caution, since, according to our audit, the originally supplied train test partition featured complete exact overlap. Therefore, the present study reports the performance using a reconstructed leakage controlled locked test set.

However, high clean-set accuracy does not necessarily ensure reliable behavior under realistic image degradation or uncertainty. Neural networks can be sensitive to common corruptions such as blur, noise, and contrast changes [15], while their confidence scores may remain poorly calibrated even when the classification accuracy is high [5]. Selective classification addresses this limitation by allowing uncertain samples to be rejected instead of forcing a prediction [8]. Grad-CAM further provides class-discriminative visual explanations that can assist in examining model decisions and failure cases [9].

Existing aquatic-plant studies have mainly emphasized classification, segmentation, or mapping performance. Compar-

atively less attention has been given to jointly examining dataset leakage, corruption robustness, confidence calibration, selective prediction, and explanation-driven failure behavior within a unified evaluation pipeline. This work addresses that gap by evaluating aquatic plant recognition from a reliability-aware perspective, integrating data-integrity control with clean performance, robustness, calibrated uncertainty, abstention behavior, and Grad-CAM-based residual failure analysis.

TABLE I
COMPARISON WITH EXISTING WORK USING THE SAME DATASET

Study	Data Split	Task	Acc.	Reliability Evaluation
Rahman et al. [4]	1062/406 supplied	Multi-task classification + segmentation	98.77%	Grad-CAM / Grad-CAM++
This study	744/161/157 reconstructed	14-class classification	96.18%	Robustness + Calibration + Selective Prediction + Cross-CAM

Note: Results are not directly comparable because this study uses a reconstructed leakage-controlled locked test set, whereas Rahman et al. used the originally supplied partition.

Rahman et al. [4] evaluated the same dataset using the originally supplied partition and reported 98.77% accuracy. However, our integrity audit identified exact counterparts for all 406 supplied test images within the training partition. Therefore, the reported values are not directly comparable. The present study instead uses a reconstructed leakage-controlled split and extends evaluation to robustness, calibration, selective prediction, and Cross-CAM analysis.

III. METHODOLOGY

Fig. 1 summarizes the overall workflow of the proposed study, starting from dataset integrity auditing and group-aware data reconstruction, followed by model selection and fine-tuning, and concluding with locked-test reliability evaluation, explainability analysis, and integrated failure analysis.

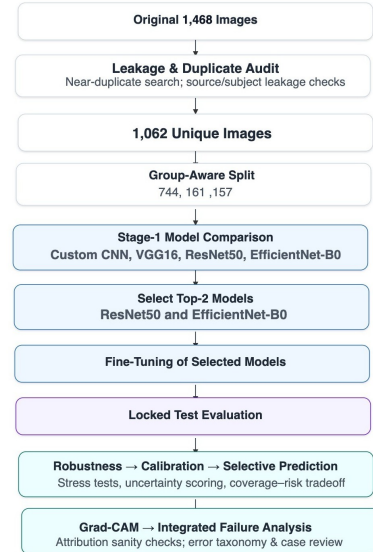


Fig. 1. Overall framework of the proposed reliability-aware aquatic plant classification pipeline, including dataset integrity auditing, group-aware splitting, model selection, locked-test evaluation, robustness analysis, calibration, selective prediction, Grad-CAM explainability, and integrated failure analysis.

A. Dataset Integrity and Preparation

The study used a UAV-based aquatic plant classification dataset that contains 1,468 images from 14 classes [7]. The supplied train–test partition was audited before model development for data leakage. Exact duplicates of images were identified using MD5 hashing of the binary file content. For two images x_i and x_j , the duplicate decision was defined as

$$D_{\text{exact}}(x_i, x_j) = \begin{cases} 1, & H_{\text{MD5}}(x_i) = H_{\text{MD5}}(x_j), \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $H_{\text{MD5}}(\cdot)$ denotes the MD5 hash. This computed from the binary content of an image file. The audit revealed that all 406 images in the supplied test set had exact match in the training set. Consequently, the original test partition was excluded from evaluation. After removing duplicates, 1062 unique images remained.

Perceptual hashing algorithm used to further control visually redundant samples. This generated near-duplicate candidates using a Hamming-distance threshold of 4. Structural similarity (SSIM) [16] was then applied for confirmation. Pairs having $\text{SSIM} \geq 0.98$ treated as near-duplicates. This process produced 77 candidate pairs and among them 60 were confirmed. Confirmed near-duplicate images were grouped so that related samples were kept within the same data partition. A group-aware stratified split was then created using approximately 70% training, 15% validation, and 15% testing, resulting in 744, 161, and 157 images, respectively. The final test set was locked and was not used during model training, hyperparameter selection, or calibration fitting.

B. Preprocessing and Data Augmentation

All images were resized to 224×224 pixels. ImageNet normalization was applied using mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$. Training images were augmented using random rotation up to $\pm 10^\circ$, random resized cropping with scale range 0.85–1.0, horizontal flipping, and mild brightness, contrast, and saturation variation. Validation and test images used only deterministic resizing and normalization to ensure consistent evaluation. A batch size of 32 was used throughout the experiments, and all models were trained using a fixed random seed of 42; run-to-run variability was not characterized in this study.

C. Model Development and Selection

Four convolutional architectures were evaluated: a custom CNN, VGG16 [11], ResNet50 [12], and EfficientNet-B0 [13]. The custom CNN was trained from scratch using four convolutional blocks with 32, 64, 128, and 256 filters, followed by adaptive average pooling and fully connected layers. The transfer-learning models were initialized using ImageNet-pretrained weights [14]; their original classification heads were replaced with 14-class output layers.

During the initial comparison, pretrained feature extractors were frozen and only the classification heads were optimized. Cross-entropy loss and AdamW optimization [17] were used,

together with learning-rate reduction and validation-based early stopping. Training was limited to a maximum of 100 epochs, with early stopping using a patience of 10 epochs. Model selection was based on validation macro-F1 rather than test performance. ResNet50 and EfficientNet-B0 achieved the two highest validation macro-F1 scores and were therefore selected for subsequent fine-tuning and reliability analysis. Fine-tuning selectively unfroze upper feature layers while applying smaller learning rates to pretrained parameters and a higher learning rate to the classifier. The checkpoint with the highest validation macro-F1 was retained for final evaluation.

D. Reliability-Aware Evaluation

The selected fine-tuned models were first evaluated once on the locked test set using accuracy, macro-precision, macro-recall, macro-F1, weighted-F1, and one-vs-rest ROC-AUC. Reliability was then assessed from four complementary perspectives.

1) *Corruption robustness*: Following the motivation of common-corruption testing [15], six image degradations were applied: Gaussian blur, Gaussian noise, brightness variation, contrast reduction, JPEG compression, and occlusion. The three severity levels are summarized in Table ?? . For stochastic corruptions, the same noise realization or occlusion location was retained across severity levels while only the magnitude was increased. No corrupted images were used for retraining or model selection.

The selected severity levels were defined as controlled and reproducible stress tests rather than physical models of a specific UAV platform. They approximate progressively stronger aerial-image degradations caused by blur, sensor noise, illumination variation, compression, and partial occlusion.

2) *Calibration and uncertainty*: Model confidence was evaluated using expected calibration error (ECE), negative log-likelihood (NLL), Brier score, maximum softmax confidence, and predictive entropy. Temperature scaling [5] was fitted using validation logits only and was then applied to the locked test predictions. Because temperature scaling does not alter the predicted class label, changes in calibration could be studied without changing accuracy or macro-F1.

3) *Selective prediction*: Selective classification [8] was used to evaluate whether uncertain predictions could be rejected rather than automatically accepted. Test samples were ranked using calibrated confidence and predictive entropy. Coverage, selective accuracy, selective risk, area under the risk–coverage curve (AURC), excess AURC, and error-rejection rate were measured at several fixed coverage levels. No coverage threshold was optimized using the locked test set.

4) *Explainability and failure analysis*: Grad-CAM [9] was applied to `layer4[-1]` of ResNet50 and `features[-1]` of EfficientNet-B0. For a target class c , Grad-CAM estimates the importance of each feature map using the gradient-based weight:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad (2)$$

where A^k denotes the k -th activation map, y^c represents the class score, and Z is the number of spatial locations. The final Grad-CAM localization map is computed as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (3)$$

For complementary post-hoc explanation, Score-CAM [18] was applied to the selected ResNet50 model on representative locked-test cases. ResNet50 was chosen because it achieved the strongest overall reliability performance, while Score-CAM was limited to this model due to its higher computational cost. Unlike Grad-CAM, Score-CAM uses activation-based importance without gradient information. The importance weight for each activation map is obtained as:

$$w_k^c = \frac{\exp(s^c(X_k))}{\sum_j \exp(s^c(X_j))}, \quad (4)$$

where X_k represents the input masked by the k -th activation map and $s^c(\cdot)$ denotes the model confidence score for class c . The Score-CAM localization map is then generated as:

$$L_{\text{Score-CAM}}^c = \text{ReLU} \left(\sum_k w_k^c A^k \right). \quad (5)$$

The two CAM methods were compared qualitatively to examine differences in model attention across correct, uncertain, and erroneous predictions.

IV. RESULTS AND DISCUSSION

A. Dataset Audit and Model Selection

The integrity audit showed that all 406 images in the test partition originally supplied had exact counterparts in the training partition, making the original split unsuitable for unbiased evaluation. After removing exact duplicates, 1,062 unique images remained. Perceptual-hash screening generated 77 near-duplicate candidates, of which 60 pairs were confirmed using SSIM. Group-aware repartitioning produced 744 training, 161 validation, and 157 locked test images with no identified exact or confirmed near-duplicate leakage across partitions.

Table II summarizes validation-based architecture selection and the final locked-test performance of the selected models. ResNet50 achieved the highest macro-F1 validation of 98.13%, followed by EfficientNet-B0 with 97.52%. VGG16 reached 97.50%, while custom CNN achieved 96.89%. Therefore, ResNet50 and EfficientNet-B0 were selected before locked test set accessed for final evaluation.

B. Locked-Test Classification Performance

In the leakage-controlled locked test set, ResNet50 correctly classified 151 of 157 images (96.18%; 95% Wilson CI: 91.91–98.24%) with a macro-F1 of 95.90%. EfficientNet-B0 correctly classified 150 of 157 images (95.54%; 95% Wilson CI: 91.08–97.82%) with a macro-F1 of 95.21%. The clean-set

TABLE II
VALIDATION-BASED MODEL SELECTION AND LOCKED-TEST PERFORMANCE

Model	Val. F1 (%)	Selected	Test Acc. (%)	Test F1 (%)	AUC
Custom CNN	96.89	No	93.24	92.86	0.9901
VGG16	97.50	No	94.67	94.12	0.9953
EfficientNet-B0	97.52	Yes	95.54	95.21	0.9974
ResNet50	98.13	Yes	96.18	95.90	0.9995

Note: Model selection was performed based on validation macro-F1. The final evaluation was conducted on the locked test set.

difference between the models was small. This motivates further comparison through robustness, calibration, and selective-prediction analyses.

The class-level prediction pattern of the final ResNet50 model is shown in Fig. 2. The matrix confirms 151 correct predictions and six errors in the 157-image locked test set.

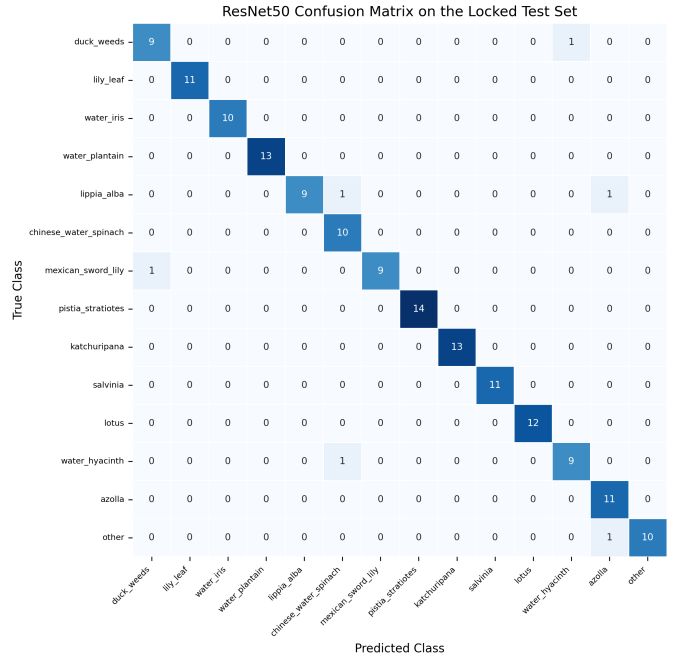


Fig. 2. Confusion matrix of the fine-tuned ResNet50 on the 157-image locked test set. Rows denote true classes and columns denote predicted classes.

Fine-tuning improved EfficientNet-B0 substantially over its frozen counterpart by increasing the test macro-F1 from 92.66% to 95.21%.

C. Robustness Under Image Corruption

The two models showed clearer differences under evaluation of corrupted inputs. ResNet50 achieved a mean corrupted macro-F1 of 92.80%, corresponding to an average drop of only 3.09 percentage points from clean performance across all corruption types and severity levels. EfficientNet-B0 achieved 90.62%, with a larger average reduction of 4.58 percentage points.

At the highest severity level, ResNet50 retained a mean of 91.52% macro-F1, where EfficientNet-B0 decreased to

83.40%. ResNet50’s most difficult condition was severe Gaussian blur, where macro-F1 fell to 89.44%. EfficientNet-B0 was particularly sensitive to severe contrast reduction, reaching only 64.20% macro-F1. These results show that similar clean-set performance can conceal substantially different robustness characteristics.

D. Calibration and Selective Prediction

Temperature scaling [5] was fitted using validation logits only to improve probability calibration without changing predicted class labels. For a logit vector $\mathbf{z}$, the calibrated class probability was computed as

$$p_k = \frac{\exp(z_k/T)}{\sum_{j=1}^K \exp(z_j/T)}, \quad (6)$$

where $T > 0$ is the temperature parameter learned and $K = 14$ is the number of classes. For ResNet50, expected calibration error decreased from 0.103 to 0.030, while NLL decreased from 0.2158 to 0.1532 and Brier score from 0.0859 to 0.0667. Its confidence gap also improved from -9.31 percentage points to approximately -0.05 percentage points. EfficientNet-B0 showed a smaller improvement, with ECE decreasing from 0.057 to 0.043.

The selective prediction further demonstrated the practical value of calibrated uncertainty. With approximately 90% coverage, ResNet50 achieved 99.30% selective accuracy while rejecting 5 of its 6 original test errors (83.33%). EfficientNet-B0 achieved 97.89% selective accuracy and rejected 4 of its 7 original errors (57.14%) at a comparable coverage level. ResNet50 also obtained a lower confidence-based AURC of 0.0034 compared with 0.0059 for EfficientNet-B0, indicating better uncertainty in this evaluation.

TABLE III
RELIABILITY RESULTS OF THE FINAL MODELS

Model	Mean Corr. F1 (%)	Strong F1 (%)	ECE B/A	Sel. Acc. (%)	Err. Rej. (%)
EfficientNet-B0	90.62	83.40	0.057/0.043	97.89	57.14
ResNet50	92.80	91.52	0.103/ 0.030	99.30	83.33

Note: Strong F1 denotes the mean macro-F1 at corruption severity level 3. B/A denotes before/after temperature scaling. Selective accuracy and error rejection are reported at approximately 90% coverage. Error rejection denotes the proportion of original incorrect predictions excluded by the rejection mechanism.

The reliability differences between the final models are further illustrated in Fig. 3. ResNet50 retained stronger performance under severe corruption and left fewer residual errors as selective coverage was reduced.

E. Explainability and Integrated Failure Analysis

Representative Grad-CAM and Score-CAM explanations from ResNet50 are shown in Fig. 4. The selected cases include high-confidence correct predictions, low-confidence correct predictions, and high-confidence errors, allowing model attention to be examined under different reliability conditions.

Grad-CAM and Score-CAM analysis qualitatively showed that many correctly classified samples produced activation over

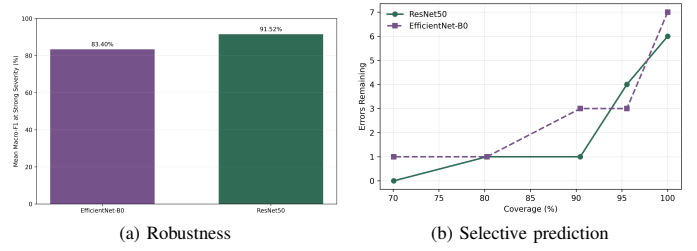


Fig. 3. Reliability comparison of the final models: (a) strong-corruption macro-F1 and (b) residual errors under selective prediction.

Cross-CAM Comparison for Representative ResNet50 Predictions

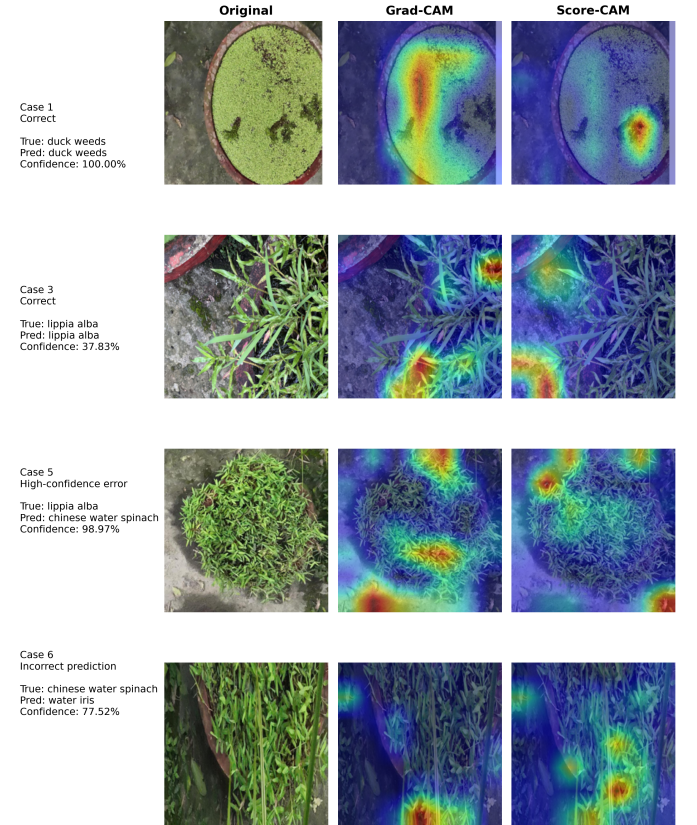


Fig. 4. Cross-CAM comparison of Grad-CAM and Score-CAM explanations for representative ResNet50 predictions across different confidence and error conditions.

visually salient plant regions, whereas several uncertain or erroneous cases exhibited more diffuse or competing attention patterns. Importantly, the explainability analysis was linked with confidence and uncertainty rather than being treated as an isolated visualization.

The integrated failure analysis identified 13 model-error records: six from ResNet50 and seven from EfficientNet-B0. Three locked-test images were misclassified by both architectures, suggesting that some samples represented common difficult cases rather than purely architecture-specific failures. A notable case-level confusion occurred from *Lippia alba* to Chinese water spinach, appearing in three model-error records across two locked-test images and affecting both

models. The mean calibrated confidence for these records was approximately 98.9%, with a maximum of 99.9%. Because this observation is based on only two locked-test images, it should be interpreted as an indicative failure case rather than a generalizable class-level confusion. The corresponding ResNet50 error is visible in Fig. 2.

ResNet50 produced only one error with calibrated confidence above 95%, whereas EfficientNet-B0 produced three. This observation complements the corruption, calibration, and selective-prediction results; in this evaluation, ResNet50 showed the stronger overall reliability profile. Nevertheless, the persistence of shared high-confidence errors shows that calibration and abstention cannot eliminate every visually ambiguous case. Consequently, reliable UAV-based aquatic plant recognition should be assessed through a combination of clean performance, robustness, uncertainty quality, rejection behavior, and explanation-based failure analysis rather than classification accuracy alone.

V. LIMITATIONS

The study has several limitations. First, the data consist of only 1062 unique images after removing duplicates, which limits the variability of the data for model training and assessment. Second, the model was trained using one random seed (42), so the variance of the results was not estimated for different runs. Third, while the reconstructed split was tested for exact and confirmed near duplicates, other types of similarities could have been present. Fourth, robustness was evaluated for six types of image corruption, which might be insufficient for all UAV conditions. Fifth, the locked test set is relatively small, thus limiting the estimations for class specific and selective prediction performance. Temperature scaling also improved global calibration, but it cannot ensure accurate calibration for each individual prediction. Finally, Grad-CAM and Score-CAM provide qualitative visual explanations rather than causal evidence, and localization accuracy was not quantitatively evaluated.

VI. CONCLUSION

This study proposed a reliability aware transfer learning framework for UAV-based aquatic plant classification whose core contributions are concerned with data integrity, robustness, calibration, selective prediction, and explainability. First, the analysis revealed that the initial dataset has severe cross partition duplication, which prompted this study to reconstruct a leakage-controlled train-validation-test paradigm. On the 157 images locked test set, ResNet50 correctly classified 151 images, achieving 96.18% accuracy and 95.90% macro-F1. In this test set, ResNet50 outperformed EfficientNet-B0 in terms of robustness and related uncertainty behaviors. Improved confidence calibrated the confidence of the model and selective prediction increased reliability by achieving 99.30% selective accuracy with around 90% coverage and rejecting five out of six original test errors (83.33%). In addition, the confusion matrix, Grad-CAM and Score-CAM analysis, and integrated

failure analysis demonstrated that a small number of difficult and high confidence errors remained despite a strong overall performance. These findings highlight that reliable UAV-based aquatic plant recognition should not be assessed only by cleaning set accuracy but through a broader evaluation of dataset integrity, robustness, calibrated confidence, abstention behavior, explainability, and residual failure patterns.

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